metals

Purpose

Heavy metals listed in constituents.cs can be routed through the network as loads. In 62.0.1 they enter the model only through export coefficients (exco_hmet.exc) and delivery ratios (dr_hmet.del) and are routed with the water. There is no metal property database, and soils, plants, channels and reservoirs have no metal processes.

file.cio has slots for four metal files:

File	Named in	Status in 62.0.1
metals.mtl	hru_parm_db line of file.cio	No reader. The file is not opened.
hmet_hru.ini	init line of file.cio	Read, but the values are not used.
hmet_water.ini	init line of file.cio	No reader. The file is not opened.
exco_hmet.exc, dr_hmet.del	export coefficient and delivery ratio lines of file.cio	Read. Metal loads from routing objects.

Set metals.mtl, hmet_hru.ini and hmet_water.ini to null in file.cio unless a later revision uses them. The hmet columns of initial.cha and initial.aqu name records of these files and have no effect.

hmet_hru.ini

Initial heavy metals in the soil and on plants of HRUs. The reader expects this layout:

- Line 1: title.
- For each record: a header line, a name line, then for each metal in the order of constituents.cs two lines, label soil and then label plant.

The counting pass expects one line per metal instead of two, so files with more than one record are miscounted. The values (soil and plant concentrations in mg/kg) are stored but no part of the model uses them.

Related files

- constituents.cs: the list of metals.
- file.cio: the file name slots.
- Pesticides and pathogens: the constituents with full process models.

Sources

- Reader: hmet_hru_aqu_read.f90, called from proc_read.f90
- Type: type cs_soil_init_concentrations (hmet_soil_ini) in constituent_mass_module.f90
- File name slots: in_parmdb%hmetcom_db, in_init%hmet_soil, in_init%hmet_water in input_file_module.f90
- Metal list: constit_db_read.f90
- Metal loads: exco_read_hmet.f90, dr_read_hmet.f90

Item	Fortran variable
hmet_hru.ini record name	hmet_soil_ini%name
soil value	hmet_soil_ini%soil
plant value	hmet_soil_ini%plt
metals.mtl slot	in_parmdb%hmetcom_db

Item	Fortran variable